

A Machine Learning Pipeline for Toxicity Prediction in Tox21: Integrating Molecular Descriptors and Structural Fingerprints

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Abstract—Aim: This study aims to develop and evaluate a robust machine learning pipeline for binary toxicity classification, focusing on the NR-AhR (Aryl Hydrocarbon Receptor) pathway using the Tox21 dataset.

Methods: Several classifiers, including Logistic Regression, Support Vector Machines (SVM), and Random Forest, were trained on different molecular representations: 2D molecular descriptors, Morgan fingerprints, and a combined feature set. The pipeline included data pre-processing, feature selection (Mutual Information and Recursive Feature Elimination), hyperparameter optimization via GridSearchCV, and model evaluation using stratified 5-fold cross-validation.

Results: Combining physicochemical descriptors with structural fingerprints consistently improved predictive performance. The best-performing model, a Logistic Regression trained on mixed features, achieved a Precision-Recall AUC of 0.721. While ROC-AUC values exceeded 0.90 across models, this metric was less informative due to the strong class imbalance, making PR-AUC a more reliable indicator of performance.

Conclusion: The proposed pipeline demonstrates that integrating complementary molecular representations enhances toxicity prediction. The results also emphasize the importance of using evaluation metrics, such as PR-AUC, that are appropriate for imbalanced datasets.

Index Terms—Toxicity prediction, Tox21, Machine Learning, Molecular Descriptors, Morgan Fingerprints, Cheminformatics, NR-AhR.

1 INTRODUCTION

THE assessment of chemical toxicity is a fundamental pillar of public health and environmental protection. With thousands of new compounds synthesized annually, traditional *in vivo* toxicological testing—which is ethically sensitive, costly, and time-consuming—is no longer sufficient to meet regulatory demands. The Toxicology in the 21st Century (Tox21) initiative was launched to address this gap by leveraging high-throughput screening (HTS) to evaluate the biological activity of small molecules across a wide range of pathways.

1.1 Biological Background: The AhR Pathway

The Aryl Hydrocarbon Receptor (AhR) is a ligand-activated transcription factor that plays a crucial role in the detection of and adaptation to chemical exposure. It is particularly

known for mediating the toxicity of environmental pollutants such as dioxins. Activation of the NR-AhR pathway can lead to significant biological changes, including altered xenobiotic metabolism, immunotoxicity, and developmental defects. Predicting AhR activation is therefore a key objective in modern toxicological modeling.

1.2 Machine Learning in Cheminformatics

Machine learning (ML) has revolutionized the field of Quantitative Structure-Activity Relationship (QSAR) modeling. By encoding chemical structures into numerical vectors—descriptors and fingerprints—ML algorithms can learn complex mappings between chemical structure and biological activity. In particular, supervised learning allows for the creation of predictive models that can rapidly screen vast virtual libraries of compounds, prioritizing those with favorable safety profiles.

1.3 Study Objectives

The objective of this study is to design and evaluate a machine learning pipeline for the binary classification of compounds as active or inactive toward the NR-AhR target. Specifically, this work aims to:

- 1) Assess the impact of different molecular representations (2D descriptors, Morgan fingerprints, and their combination) on predictive performance.
- 2) Investigate feature selection strategies to reduce redundancy and improve model stability in high-dimensional cheminformatics data.
- 3) Compare the performance of linear and non-linear classifiers under a consistent evaluation framework.
- 4) Properly address class imbalance inherent to high-throughput screening (HTS) data using class weighting and suitable evaluation metrics (e.g., PR-AUC).

2 MATERIALS

The dataset employed in this study is derived from the Tox21 initiative, specifically focusing on the NR-AhR (Aryl Hydrocarbon Receptor) bioassay. This dataset consists of chemical compounds annotated based on their ability to activate the AhR signaling pathway, providing a binary classification task (active vs. inactive).

The initial dataset included 7,831 compounds. After preprocessing—removing molecules that could not be successfully parsed and discarding samples with missing labels for the selected endpoint—a final dataset of 6,542 compounds was obtained. This filtering step ensures consistent supervision during model training, at the cost of reducing the available data.

2.1 Feature Description

Chemical information was represented using two complementary feature types, capturing both global physicochemical properties and local structural patterns:

- **Molecular Descriptors (2D):** A set of 217 numerical features computed using RDKit, describing global properties of the molecules. These include physicochemical attributes (e.g., molecular weight, logP), topological descriptors, and surface-related measures such as Topological Polar Surface Area (TPSA). These features provide an interpretable representation of molecular characteristics relevant to biological activity.
- **Morgan Fingerprints:** Circular fingerprints generated with radius 2 and a dimensionality of 2048 bits. Each bit encodes the presence or absence of specific substructures derived from local atomic environments. Unlike descriptors, these features capture fine-grained structural information and are particularly effective in modeling structure–activity relationships.

In addition to using each representation independently, a combined feature set was constructed by concatenating molecular descriptors and Morgan fingerprints. This hybrid representation aims to leverage both global and local information to improve predictive performance.

2.2 Software and Libraries

All experiments were conducted in Python. Data manipulation and preprocessing were performed using `pandas` and `NumPy`. Cheminformatics operations, including molecule parsing, sanitization, and feature extraction, were carried out using the `RDKit` library.

Machine learning tasks—including feature scaling, feature selection, model training, and evaluation—were implemented using `scikit-learn`. In particular, utilities such as `StandardScaler`, `SelectKBest`, `RFE`, and `GridSearchCV` were employed to construct a consistent and reproducible pipeline.

Data visualization, including exploratory plots and performance curves (e.g., Precision-Recall and ROC curves), was performed using `matplotlib` and `seaborn`.

3 METHODS

3.1 Data Preparation and Cleaning

Raw SMILES strings were converted into RDKit molecule objects. Molecules that failed sanitization or could not be successfully parsed were discarded. In addition, samples with missing labels for the selected endpoint (NR-AhR) were removed to ensure consistent supervision.

The final dataset consists of 6,542 compounds, with 768 active (11.7%) and 5,774 inactive (88.3%) samples. This pronounced class imbalance is a critical aspect of the problem and directly influences both model training and evaluation strategy.

3.2 Exploratory Data Analysis (EDA)

Exploratory data analysis was performed to characterize the dataset, assess feature behavior, and identify potential issues such as class imbalance and feature redundancy.

3.2.1 Class Distribution

The distribution of active and inactive compounds is shown in Figure 1. The strong imbalance motivates the use of cost-sensitive learning (via class weighting) and evaluation metrics that are appropriate for imbalanced settings, such as Precision-Recall AUC (PR-AUC).

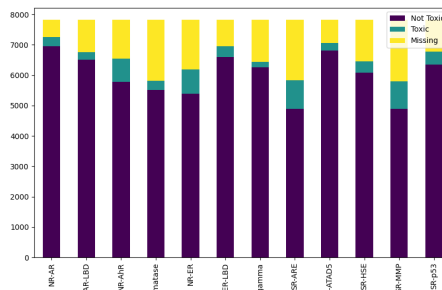


Fig. 1: Class distribution of the NR-AhR assay. Only 11.7% of compounds are labeled as active, highlighting the severe class imbalance.

3.2.2 Feature Distributions

To assess the discriminative potential of individual descriptors, features were ranked using Mutual Information (MI), and the top 10 were visualized using violin plots (Figure 2). Several features exhibit clear differences in distribution between active and inactive compounds, suggesting their relevance for classification.

It is important to note that this analysis is purely exploratory: since it is performed on the full dataset, it is not used to drive feature selection in order to avoid data leakage.

3.2.3 Correlation Analysis

To quantify linear relationships between descriptors, Pearson’s correlation coefficient was computed. The coefficient r is defined as:

$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}} \quad (1)$$

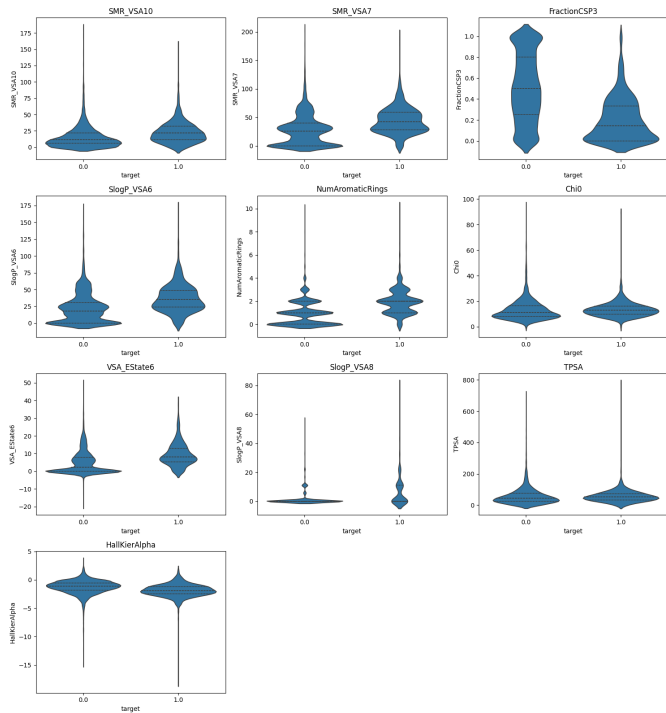


Fig. 2: Violin plots of the top 10 descriptors ranked by Mutual Information. Distinct distribution patterns between classes are observed for several features.

Due to the high dimensionality of the descriptor space, instead of visualizing the full correlation matrix, we focused on identifying highly correlated feature pairs. The analysis revealed 94 pairs of descriptors with strong correlation ($r > 0.95$), indicating substantial redundancy in the feature space and motivating the application of feature selection techniques.

3.3 Feature Engineering and Scaling

All preprocessing steps were performed exclusively on the training data to prevent information leakage. Continuous features were standardized to have zero mean and unit variance:

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

where μ and σ are estimated on the training set and subsequently applied to the test set. This transformation is essential for scale-sensitive models such as Support Vector Machines and Logistic Regression, ensuring stable optimization and comparable feature contributions.

3.4 Feature Selection Strategy

Given the high dimensionality and the presence of strong multicollinearity among descriptors, a two-stage feature selection strategy was applied. Feature selection was performed using only the training data to avoid optimistic bias.

- 1) **Univariate Selection:** Features were ranked using Mutual Information (MI), which captures both lin-

ear and non-linear dependencies between each feature X and the target Y :

$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) \quad (3)$$

This step provides a model-agnostic ranking of individual feature relevance.

- 2) **Recursive Feature Elimination (RFE):** A Logistic Regression model with regularization was used to iteratively remove less informative features based on coefficient magnitude. This step accounts for feature interactions within a multivariate setting.

The final feature set was defined as the intersection of features selected by both methods, retaining only those consistently identified as informative. This conservative strategy prioritizes robustness and reduces the risk of selecting spurious predictors, resulting in 108 descriptors.

3.5 Model Training and Evaluation

Three supervised learning algorithms were selected to cover different inductive biases:

- **Logistic Regression (LR):** A regularized linear model providing a strong and interpretable baseline.
- **Support Vector Machine (SVM):** A kernel-based model (RBF kernel) capable of capturing non-linear decision boundaries.
- **Random Forest (RF):** An ensemble of decision trees, inherently robust to noise and capable of modeling complex feature interactions.

Hyperparameters were optimized using GridSearchCV with stratified 5-fold cross-validation on the training set. Stratification ensures that class proportions are preserved across folds, which is critical given the strong class imbalance.

Model selection was primarily based on Precision-Recall AUC (PR-AUC), as it provides a more informative assessment than accuracy or ROC-AUC in imbalanced classification settings by focusing on the minority (positive) class. ROC-AUC was reported as a complementary metric but was not used as the primary optimization criterion.

Additionally, the Matthews Correlation Coefficient (MCC) was used as a threshold-dependent metric that remains reliable under class imbalance:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (4)$$

All final evaluations were performed on a held-out test set not used during model selection, providing an unbiased estimate of generalization performance.

4 RESULTS

4.1 Overall Model Performance

Models trained on the combined feature representation (molecular descriptors + Morgan fingerprints) consistently outperformed those trained on individual feature sets, confirming the complementarity between global physicochemical properties and local structural information.

TABLE 1: Model performance comparison using the mixed feature representation.

Model	PR-AUC	ROC-AUC	F1	MCC
Logistic Regression	0.721	0.918	0.610	0.574
Random Forest	0.717	0.930	0.518	0.534
SVM	0.713	0.908	0.632	0.583

Table 1 reports the performance on the held-out test set.

Across models, performance remains consistent, with all approaches achieving PR-AUC values above 0.71, indicating strong predictive capability under severe class imbalance. Logistic Regression achieves the best balance in terms of PR-AUC, while SVM shows the highest F1-score and MCC.

4.2 Performance Curves

Figure 3 shows the Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves for the best-performing model (Logistic Regression trained on the mixed feature set).

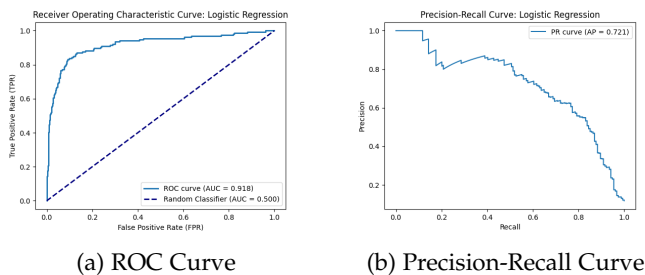


Fig. 3: Performance curves for Logistic Regression on the mixed feature representation.

While ROC curves indicate strong overall separability, the PR curve provides a more informative view of performance under class imbalance, highlighting the model’s ability to correctly identify the minority (active) class.

4.3 Confusion Matrix Analysis

The confusion matrix for Logistic Regression is shown in Figure 4. The model achieves a favorable trade-off between precision and recall, correctly identifying a substantial proportion of active compounds while maintaining a controlled number of false positives.

5 DISCUSSION AND CONCLUSIONS

This study presents a machine learning pipeline for toxicity prediction targeting the NR-AhR pathway in the Tox21 dataset. The results show that combining 2D molecular descriptors with Morgan fingerprints yields a richer and more informative representation of chemical space, effectively capturing both global physicochemical properties and local structural motifs.

5.1 Summary of Key Findings

The experimental results support the following observations:

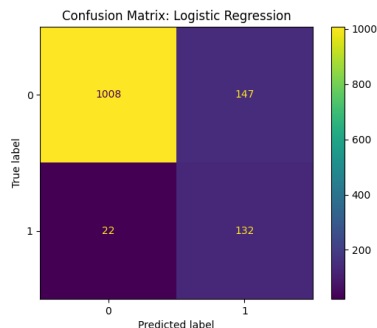


Fig. 4: Confusion matrix for Logistic Regression evaluated on the held-out test set.

- **Feature Integration:** The combination of molecular descriptors and Morgan fingerprints consistently outperforms individual feature representations, highlighting the complementary nature of global and local molecular information.
- **Model Robustness:** Competitive performance can be achieved even with relatively simple linear models such as Logistic Regression, provided that appropriate feature selection and class imbalance handling strategies are applied.
- **Evaluation Metrics:** In the context of highly imbalanced data, PR-AUC and MCC provide more reliable performance estimates than ROC-AUC, which can be overly optimistic due to the large proportion of true negatives.

5.2 Cheminformatic Interpretability

Beyond predictive performance, several selected descriptors show consistency with known chemical principles underlying AhR activation. In particular, features related to lipophilicity (e.g., $\log P$) and surface-related properties (e.g., SMR_VSA10) are aligned with the known tendency of AhR ligands to be hydrophobic molecules capable of interacting with the receptor binding pocket.

5.3 Limitations and Future Directions

This study is limited by its focus on a single biological endpoint, which restricts the generalizability of the findings across the full spectrum of Tox21 assays.

Future work will extend the proposed pipeline to all 12 toxicity targets and investigate more expressive deep learning approaches, such as Graph Convolutional Networks (GCNs), which operate directly on molecular graph representations and may reduce the need for manual feature engineering.

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